15届省赛（第二套）代码润色版

```c
#include <STC15F2K60S2.H>
#include "led.h"
#include "init.h"
#include "key.h"
#include "seg.h"
#include "onewire.h"
#include "iic.h"
//按键
idata unsigned char key_val,key_old,key_down,key_up;
idata unsigned char key_slow_down;
idata unsigned char key_long_flag;
idata unsigned int key_long_tick;
//数码管
idata unsigned char seg_pos;
pdata unsigned char seg_buf[]={10,10,10,10,10,10,10,10};
idata unsigned char seg_slow_down;
idata unsigned char seg_disp_mode=0;//0-温度界面 1-dac输出控制界面 2-校准值界面
//led
pdata unsigned char ucled[]={0,0,0,0,0,0,0,0};
idata unsigned int time1_2s;
idata unsigned int time2_2s;
idata unsigned int time_3s;
idata bit led_flash_flag;
idata unsigned char led_flash_tick;
idata bit T_up_flag;//温度上升标志位
idata bit T_down_flag;//温度下降标志位
//温度
idata unsigned char T_val;
idata unsigned char T_old;
idata unsigned int T_slow_down;
```

```
idata bit baojing_flag;
//dac
idata unsigned char dac_data=100;//0-255
idata bit dac_model=0;//0-温度控制 1-手动控制
//校准值
idata char jiaozhun_para=0;//-9 - 9
idata char jiaozhun_para_ct=0;
//锁定
idata bit lock_flag=0;//1-锁定 0-解锁
void key_proc()
{
    if(key_slow_down<10) return;
    key_slow_down=0;
    key_val=key_read();
    key_down=key_val&(key_val^key_old);
    key_up=~key_val&(key_val^key_old);
    key_old=key_val;
    if(key_down==12)
    {
        switch(seg_disp_mode)
        {
            case 0:
                seg_disp_mode=1;
                break;
            case 1:
                jiaozhun_para_ct=jiaozhun_para;
                seg_disp_mode=2;
                break;
            case 2:
                seg_disp_mode=0;
                jiaozhun_para=jiaozhun_para_ct;
                break;
        }
    }
    if(key_down==16)
    {
```

```
            switch(seg_disp_mode)
            {
                case 0:
                    if(lock_flag==0)
                        relay(1);
                    break;
                case 1:
                    if(dac_data<255) dac_data+=5;
                    break;
                case 2:
                    if(jiaozhun_para_ct<9) jiaozhun_para_ct++;
                    break;
            }
        }
        if(key_down==17)
        {
            switch(seg_disp_mode)
            {
                case 0:
                    if(lock_flag==0)
                        relay(0);
                    break;
                case 1:
                    if(dac_data>0) dac_data-=5;
                    break;
                case 2:
                    if(jiaozhun_para_ct>-9) jiaozhun_para_ct--;
                    break;
            }
        }
        if(key_down==13)
        {
            key_long_flag=1;
        }
        if(key_up==13)
        {
```

```
            if(key_long_tick>=1500)
            {
                lock_flag=!lock_flag;
            }
            else
            {
                dac_model=!dac_model;
            }
            key_long_flag=0;
            key_long_tick=0;
        }
    }


    void seg_proc()
    {
        if(seg_slow_down<80) return;
        seg_slow_down=0;
        switch(seg_disp_mode)
        {
            case 0:
                seg_buf[0]=11;

    seg_buf[1]=seg_buf[2]=seg_buf[3]=seg_buf[4]=seg_buf[5]=10;
                seg_buf[6]=T_val/10%10;
                seg_buf[7]=T_val%10;
                break;
            case 1:
                seg_buf[0]=12;
                seg_buf[1]=seg_buf[2]=seg_buf[3]=seg_buf[4]=10;
                seg_buf[5]=(dac_data>=100)?dac_data/100%10:10;
                seg_buf[6]=(dac_data>=10)?dac_data/10%10:10;
                seg_buf[7]=dac_data%10;
                break;
            case 2:
                seg_buf[0]=13;
```

```
seg_buf[1]=seg_buf[2]=seg_buf[3]=seg_buf[4]=seg_buf[5]=10;
            if(jiaozhun_para_ct>=0)
            {
                seg_buf[6]=10;
                seg_buf[7]=jiaozhun_para_ct%10;
            }
            else
            {
                seg_buf[6]=14;
                seg_buf[7]=(-jiaozhun_para_ct)%10;
            }
            break;

    }
}

void led_proc()
{
    ucled[0]=(!dac_model);
    ucled[1]=T_up_flag;
    ucled[2]=T_down_flag;
    ucled[3]=led_flash_flag;
    ucled[7]=(!lock_flag);
}

void get_temperature()
{
    float temp_val;
    char diff;
    if(T_slow_down<300) return;
    T_slow_down=0;
    T_old=T_val;
    temp_val=rd_temperature();
    T_val=(temp_val+jiaozhun_para<0)? 0:(unsigned char)
(temp_val+jiaozhun_para);
```

```
    if(T_val>T_old)
    {
        if(T_up_flag==0)
        {
            T_up_flag=1;
            time1_2s=0;
        }
    }
    if(T_val<T_old)
    {
        if(T_down_flag==0)
        {
            T_down_flag=1;
            time2_2s=0;
        }
    }

    diff = (char)T_val - (char)T_old;
    if(diff > 1 || diff < -1)
    {
        if(baojing_flag==0)
        {
            baojing_flag=1;
            time_3s=0;
            led_flash_tick=0;
            led_flash_flag=1;
        }
    }
}

void da_proc()
{
    float temp;
    if(dac_model==0)
    {
        if(T_val>=40) temp=5;
```

```
        else if(T_val>10) temp=(1.0/10.0)*(T_val-10)+2;
        else temp=2;
        da_write((unsigned char)(temp*51));
    }
    else
    {
        da_write(dac_data);
    }
}

void Timer1_Init(void)      //1毫秒@12.000MHz
{
    AUXR &= 0xBF;           //定时器时钟12T模式
    TMOD &= 0x0F;           //设置定时器模式
    TL1 = 0x18;             //设置定时初始值
    TH1 = 0xFC;             //设置定时初始值
    TF1 = 0;                //清除TF1标志
    TR1 = 1;                //定时器1开始计时
    ET1 = 1;                //使能定时器1中断
    EA = 1;          //使能全局中断
}

void Timer1_Isr(void) interrupt 3
{
    key_slow_down++;
    seg_slow_down++;
    T_slow_down++;

    seg_disp(seg_pos,seg_buf[seg_pos]);
    seg_pos=(++seg_pos)%8;

    if(key_long_flag)
    {
        if(++key_long_tick>=1500) key_long_tick=1501;
    }
    else
```

```
        {
            key_long_tick=0;
        }
        if(T_up_flag)
        {
            if(++time1_2s==2000)
            {
                T_up_flag=0;
                time1_2s=0;
            }
        }
        if(T_down_flag)
        {
            if(++time2_2s==2000)
            {
                T_down_flag=0;
                time2_2s=0;
            }
        }
        if(baojing_flag)
        {
            if(++led_flash_tick==200)
            {
                led_flash_tick=0;
                led_flash_flag=!led_flash_flag;
            }
            if(++time_3s==3000)
            {
                time_3s=0;
                led_flash_tick=0;
                led_flash_flag=0;
                baojing_flag=0;
            }
        }
        led_disp(ucled);
    }
```

```
void main()
{
    system_init();
    while(rd_temperature()==85.0);
    Timer1_Init();
    while(1)
    {
        key_proc();
        seg_proc();
        get_temperature();
        led_proc();
        da_proc();
    }

}
```

Fence 1